HYPERPARAMETER TUNING OF ARTIFICIAL NEURAL NETWORK (ANN) AND CONVOLUTIONAL NEURAL NETWORK (CNN)

Introduction

Hyperparameter tuning is an important process in machine learning and deep learning that helps improve model performance by selecting the most suitable configuration of parameters before training. Unlike model parameters, which are learned automatically during training, hyperparameters are set by the user and directly influence the learning process. Examples include the number of neurons, learning rate, batch size, epochs, filters, kernel size, and optimizer. Proper tuning can significantly improve accuracy, reduce overfitting, and enhance model generalization.

Task 1: Hyperparameter Optimization on ANN

In this task, an Artificial Neural Network (ANN) was developed and optimized using the Breast Cancer Dataset. The dataset contains information about tumor characteristics and is widely used for binary classification problems. The objective was to classify tumors as malignant or benign based on the available features.

Data preprocessing steps included loading the dataset, handling target labels, feature scaling using StandardScaler, and splitting the dataset into training and testing sets. Feature scaling was necessary because neural networks perform better when input values are normalized to a similar range.

The ANN model consisted of an input layer, one hidden layer with a variable number of neurons, and an output layer with a sigmoid activation function for binary classification. Hyperparameter tuning was performed using three different approaches:

1. **Manual Search:**
   Different combinations of neurons, batch sizes, and epochs were manually tested. The performance of each combination was evaluated and compared. The best configuration was obtained with 32 neurons, batch size 16, and 20 epochs, achieving an accuracy of approximately 98.68%.

2. **Randomized Search CV:**
   Randomized Search Cross Validation was used to randomly sample combinations of hyperparameters from a predefined search space. Parameters such as neurons, batch size, epochs, and optimizer were tuned automatically. The best configuration used the Adam optimizer, 16 neurons, batch size 16, and 30 epochs, producing a cross-validation score of approximately 97.58%.

3. **Grid Search CV:**
   Grid Search Cross Validation exhaustively evaluated all possible combinations of selected hyperparameters. This method required more computation but provided a systematic search for the optimal configuration. The best parameters were RMSprop optimizer, 16 neurons, batch size 16, and 20 epochs, resulting in a cross-validation score of approximately 97.80%.

The final optimized ANN model achieved a test accuracy of approximately 99.12%, demonstrating excellent performance in classifying breast cancer cases. The experiment showed that hyperparameter tuning significantly improves model effectiveness and helps identify the most suitable ANN architecture.

Task 2: Hyperparameter Optimization on CNN

In the second task, a Convolutional Neural Network (CNN) was developed and optimized using the Fashion-MNIST dataset. Fashion-MNIST is a popular image classification dataset containing grayscale images of clothing items such as shirts, trousers, shoes, bags, and dresses. The dataset consists of 60,000 training images and 10,000 testing images, each having a resolution of 28×28 pixels.

Before training, the image pixel values were normalized by dividing them by 255, converting them into a range between 0 and 1. The images were then reshaped into a format suitable for CNN processing.

The CNN architecture included a convolutional layer, max-pooling layer, flatten layer, fully connected dense layer, and an output layer with softmax activation for multi-class classification. Hyperparameter optimization was performed using the following techniques:

1. **Manual Search:**
   Various combinations of filters, batch sizes, and epochs were tested manually. The best result was obtained using 32 filters, batch size 32, and 3 epochs. This configuration achieved a validation accuracy of approximately 82.17%.

2. **Randomized Search CV:**
   Randomized Search CV was used to evaluate random combinations of filters, kernel sizes, batch sizes, epochs, and optimizers. The best configuration included the Adam optimizer, 16 filters, kernel size 5, batch size 64, and 2 epochs, resulting in a cross-validation score of **approximately 60.60%.**

3. **Grid Search CV:**
   Grid Search CV systematically explored all selected parameter combinations. The best configuration consisted of the Adam optimizer, 16 filters, kernel size 5, batch size 32, and 3 epochs. This setup achieved a cross-validation score of approximately 70.22%.

The final optimized CNN model achieved a test accuracy of approximately 72.10% on the Fashion-MNIST dataset. The experiment highlighted the impact of hyperparameters such as filters, kernel size, optimizer, batch size, and epochs on image classification performance.

Conclusion

The objective of this experiment was to perform hyperparameter optimization on both ANN and CNN models using Manual Search, Randomized Search CV, and Grid Search CV techniques. For the ANN model trained on the Breast Cancer Dataset, hyperparameter tuning improved classification performance and resulted in a test accuracy of 99.12%. For the CNN model trained on the Fashion-MNIST dataset, tuning helped identify the most effective architecture and achieved a test accuracy of 72.10%.

The study demonstrated that hyperparameter optimization is an essential step in deep learning model development. It helps improve prediction accuracy, enhances model generalization, and enables the selection of the most suitable configuration for a given problem. Among the tuning methods, Grid Search CV provides a systematic search, Randomized Search CV reduces computational cost, and Manual Search offers a simple way to understand parameter effects. Overall, hyperparameter tuning played a crucial role in achieving better performance for both ANN and CNN models.